



UG

DEPARTMENT OF
PHYSICS

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Introduction to Nanomagnetism

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Why magnetism matters?

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DE MAEZTU

Topics

Nanoscale magnetism: basic concepts.

- **Magnetic moments**
- **Magnetic interactions**
- **Magnetic anisotropy**
- **Spin orbit coupling**

Magnetism: A brief history

Magnetism is a very fascinating phenomenon described by Thales of Miletus (634-546 B.C.) who observed the attraction of Fe by Magnetite (Fe_3O_4)



Some early magnetic devices: the 'South pointer' used for orientation in China around the beginning of the present era, and a Portuguese mariner's compass from the fifteenth century.

1820 Hans-Christian Oersted discovered the true connection between electricity and magnetism

1821 Michael Faraday discovers electromagnetic induction and demonstrated the principle of the electric motor with a steel magnet

Magnetism: A brief history

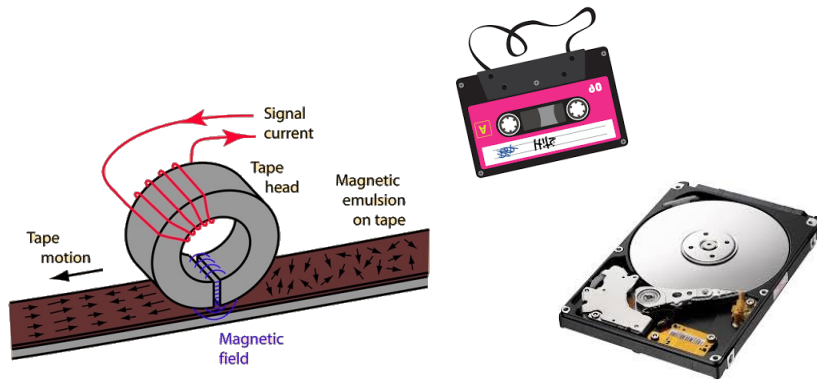
- 1845** Discovery of a connection between magnetism and light followed with the magneto-optic Faraday effect
- 1864** James Clerk Maxwell: unified theory of electricity, magnetism and light (his famous equations!)
- 1925-28** Development of Quantum Mechanics and the idea of 'spinning electron' (Uhlenbeck and Goudsmit, 1925) and relativistic quantum mechanics (Dirac, 1928).

Magnetism is **quantum** in origin and **cooperative** in character — and the glue is **electrostatic**.

It tooks more than 2500 years to understand why some rocks are magnetic, i.e. the microscopic origin of magnetism

Magnetism: relevance in science and technology

Recording technology

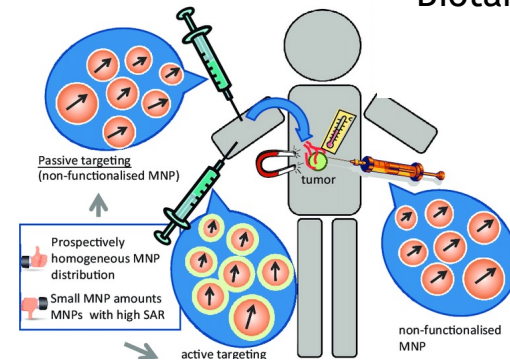


Industrial machine engineering (MAGLev)



Biomedical applications

Biotargeting

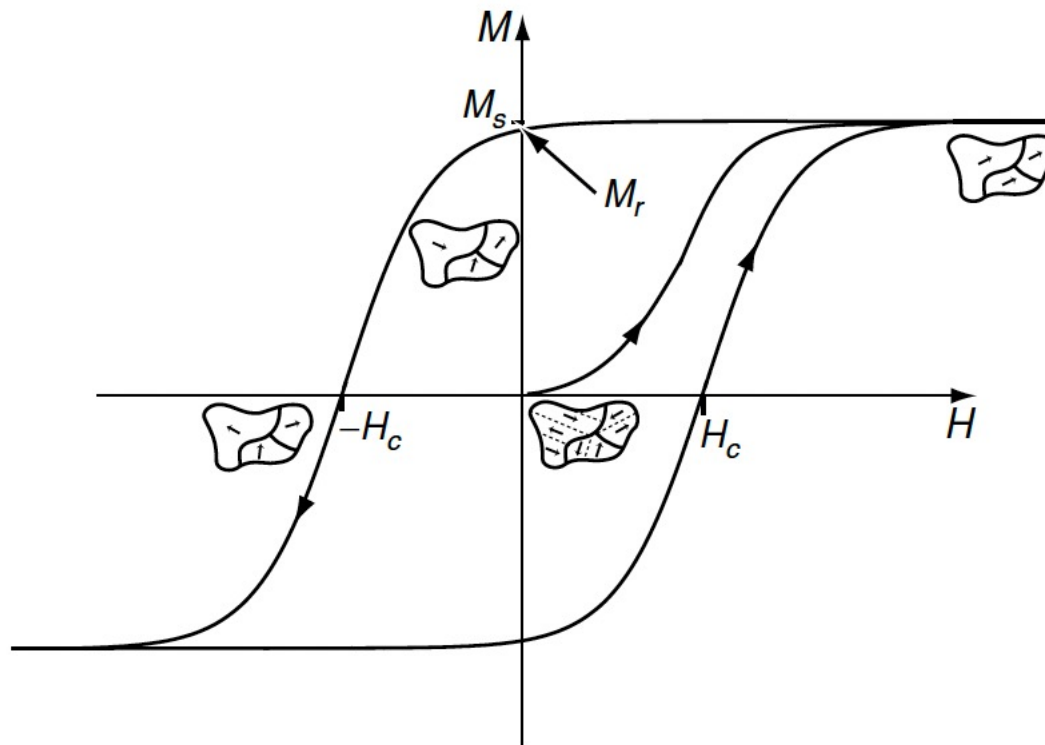


Magnetic resonance imaging



Magnetism and hysteresis

The most striking manifestation of magnetism in solids is the spontaneous magnetization of ferromagnetic materials such as iron or magnetite. This is associated with hysteresis (James Ewing, 1881).



The magnetic hierarchy

- Atomic scale: electrons carry spin and orbital moments.
- Lattice scale: exchange couples neighbouring moments.
- Mesoscale: domains minimize competing energies.
- Device scale: hysteresis determines function and loss.



*Each scale has its own language,
but they all describe the same magnetization.*

The magnetic hierarchy

- Atomic scale: electrons carry spin and orbital moments.
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Each scale has its own language, but they all describe the same magnetization.

Basic ingredients

Magnetic moment

Magnetic interactions between magnetic moments

Consequences

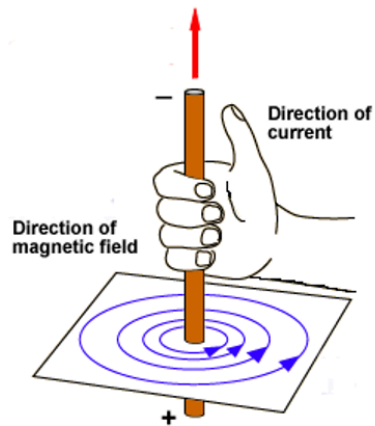
Magnetic anisotropy

Magnetic ordering

Magnetization and magnetic moment

Electrodynamics

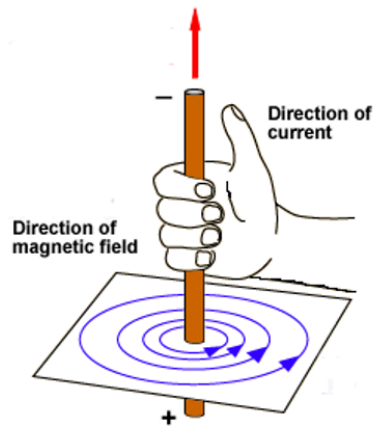
Induction of magnetic moments through moving charges



Magnetization and magnetic moment

Electrodynamics

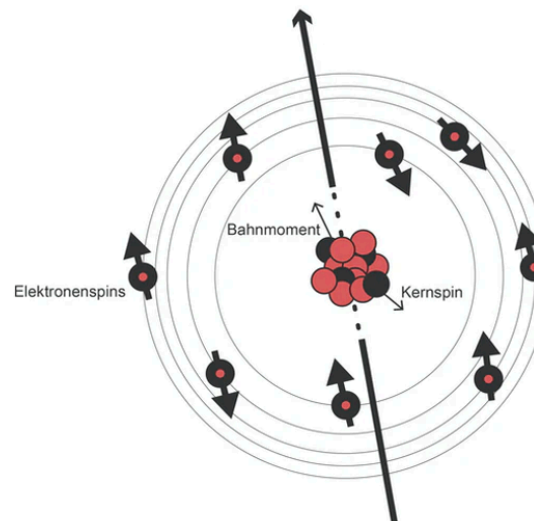
Induction of magnetic moments through moving charges



Atomic perspective

Magnetic dipole moment

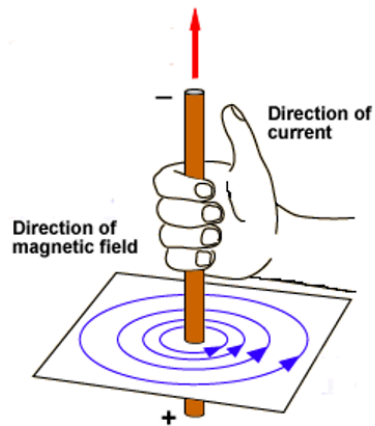
Nucleus + Electron



Magnetization and magnetic moment

Electrodynamics

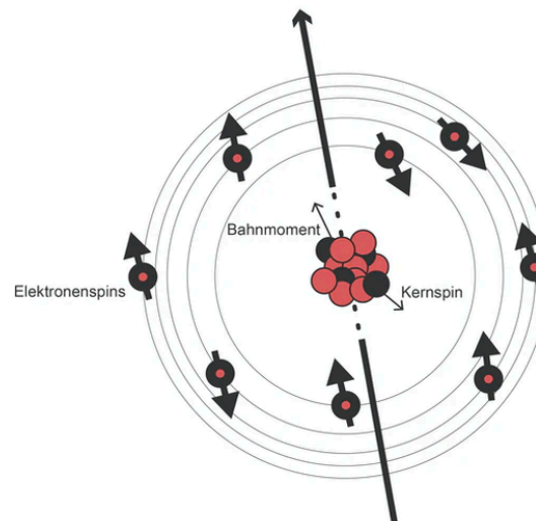
Induction of magnetic moments through moving charges



Atomic perspective

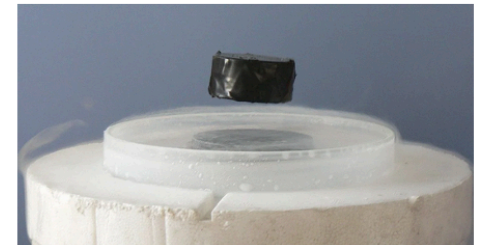
Magnetic dipole moment

Nucleus + Electron



Material properties

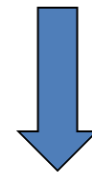
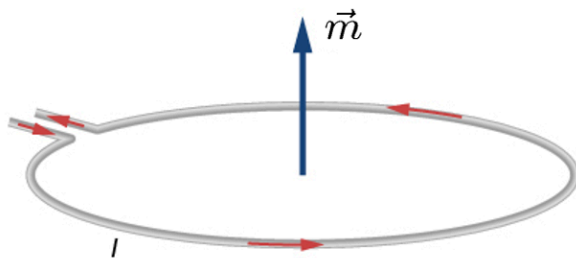
- Susceptibility
- Ferro- Para- Diamagnetism



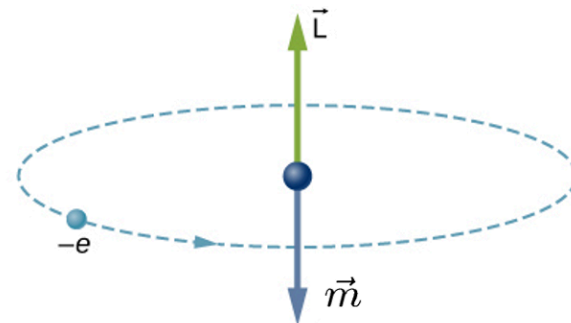
Magnetization and magnetic moment



Free currents in a ring geometry



Atomic currents



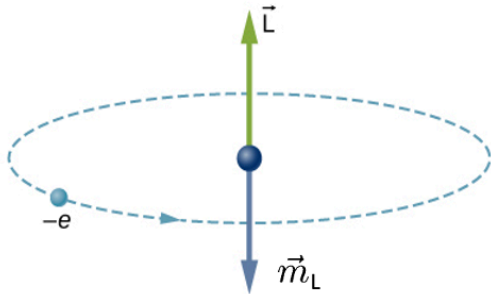
Ring currents = Orbiting electron around nucleus

Magnetic moment produced by ring currents = **“Orbital”** moment of moving electron

Atomistic magnetic moment

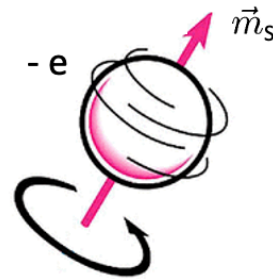
Electron **motion**

Orbital angular momentum
(m_L)



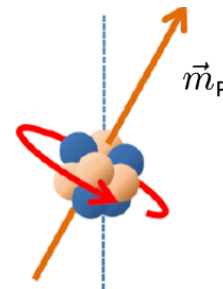
Rotation on the own axis

Spin angular momentum
(electron spin m_s)



Rotation on the own axis

Spin angular momentum
(nuclear spin m_p)



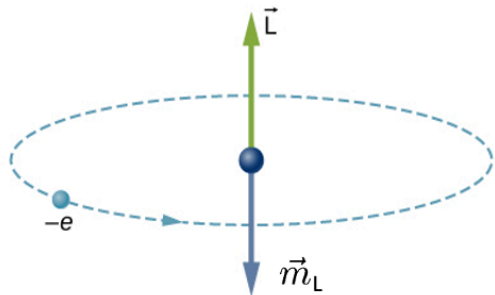
$$\vec{m}_a = \vec{m}_s + \vec{m}_L + \vec{m}_p$$

$$\vec{m}_s > \vec{m}_L \gg \vec{m}_p$$

Atomistic magnetic moment

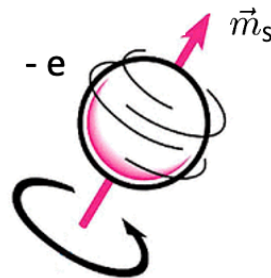
Electron **motion**

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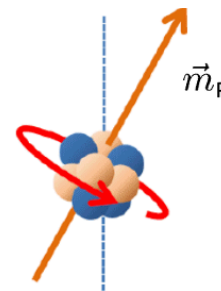
Rotation on the own axis

Spin angular momentum
(electron spin m_s)



Rotation on the own axis

Spin angular momentum
(nuclear spin m_p)



$$\vec{m}_a = \vec{m}_s + \vec{m}_L + \vec{m}_p$$

$$\vec{m}_s > \vec{m}_L \gg \vec{m}_p$$

Atomic magnetic moment: SPIN + Orbital

The magnetic moment



$$\mu = -\mu_B(L + g_e S)$$

L Orbital angular momentum

S Spin angular momentum

$g_e = 2.0023$ (Landé factor for the free electron)

μ_B Bohr Magneton

THE PUZZLE

Two facts that demand an explanation

① Iron orders at a red heat

$$T_C^{\text{Fe}} = 1043 \text{ K}$$

A robust, room-temperature alignment of $\sim 10^{23}$ spins. Something holds them parallel against fierce thermal disorder.

So how strong is the magnetic force between two neighbouring moments?

② ...but dipole-dipole coupling is tiny

$$E_{dd} \sim \frac{\mu_0 \mu_B^2}{4\pi r^3} \approx k_B \times 2 \text{ K}$$

Off by a factor of **~500**.

Rest *mass* 511 keV $\gg$ Coulomb 13.6 eV $\gg$ magnetic 0.18 meV $\approx$ 2 K.

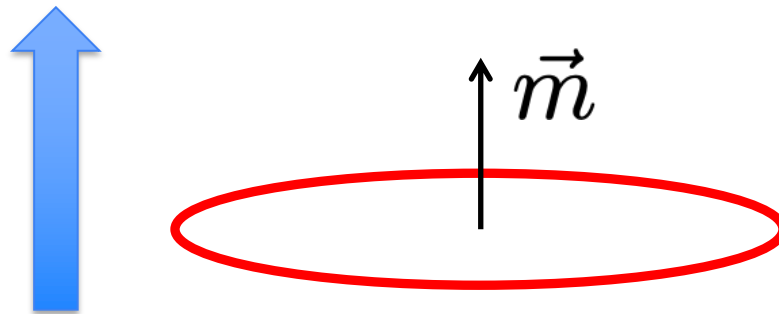
So, the aligning force is NOT magnetic. We need a different –and far stronger– mechanism.

Effect of magnetic field on magnetic moment

$$U = -\vec{m} \cdot \vec{B}$$

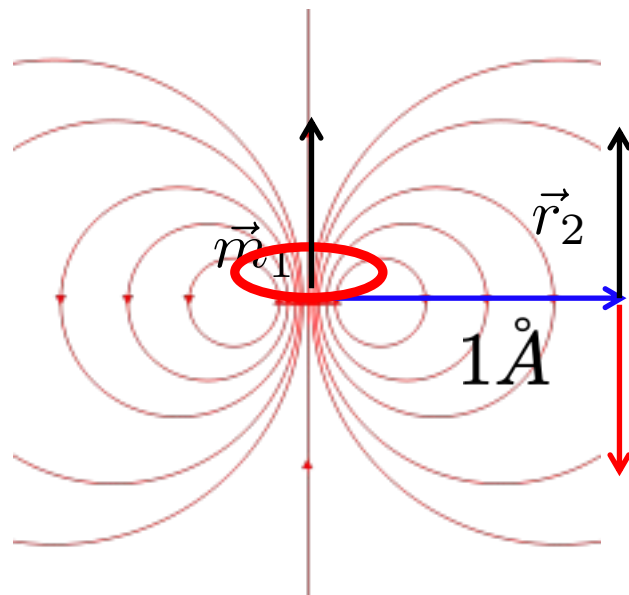
Classical formula

$\vec{B}$



Magnetic interaction

$$\vec{B}_1(\vec{r}_2) = \frac{\mu_0}{4\pi} \left(\frac{3\vec{r}_2(\vec{m}_1 \cdot \vec{r}_2)}{r^5} - \frac{\vec{m}_1}{r^3} \right)$$


 $\vec{m}_2$

$$U = -\vec{m}_2 \cdot \vec{B}_1(\vec{r}_2)$$

$$|B(1\text{\AA})| \simeq 1T$$

$$\mu_B = 58 \mu\text{eV}/T$$

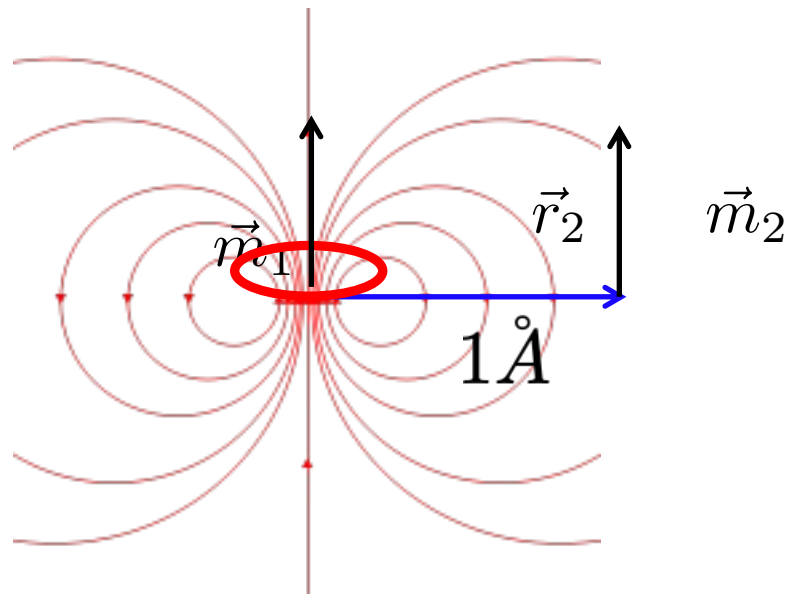
$$U_{\uparrow} \simeq 0.06 \text{ meV}$$

$$U_{\downarrow} \simeq -0.06 \text{ meV}$$

$$\vec{m}_1 = \vec{m}_2 = \mu_B(0, 0, 1)$$

Magnetic vs Coulomb interaction

Magnetic dipole-dipole



$$\vec{m}_1 = \vec{m}_2 = \mu_B(0, 0, 1)$$

$$\Delta U \simeq 0.12 \text{ meV}$$

Coulomb:

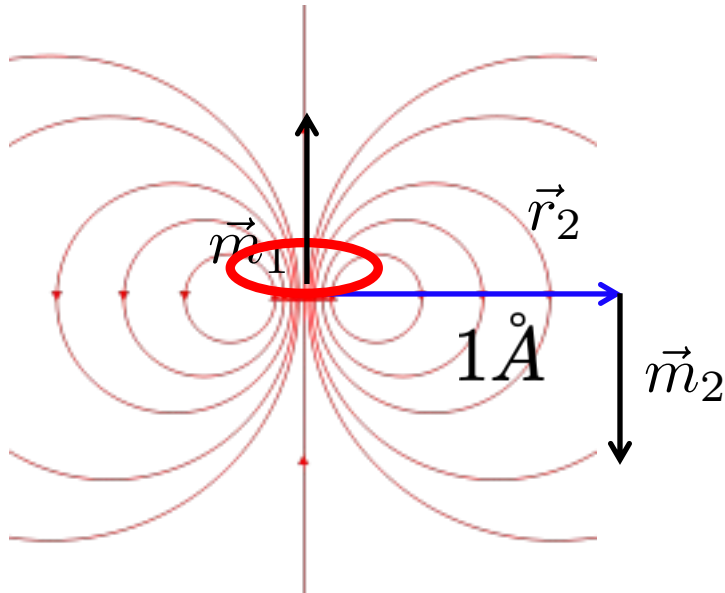
$$V(r) = \frac{q^2}{4\pi\epsilon_0 r}$$



$$V_{qq} = 13 \text{ eV}$$

Magnetic Dipolar interaction is 10^5 times weaker (at 1Å)

Other 'magnetic glue' must exist



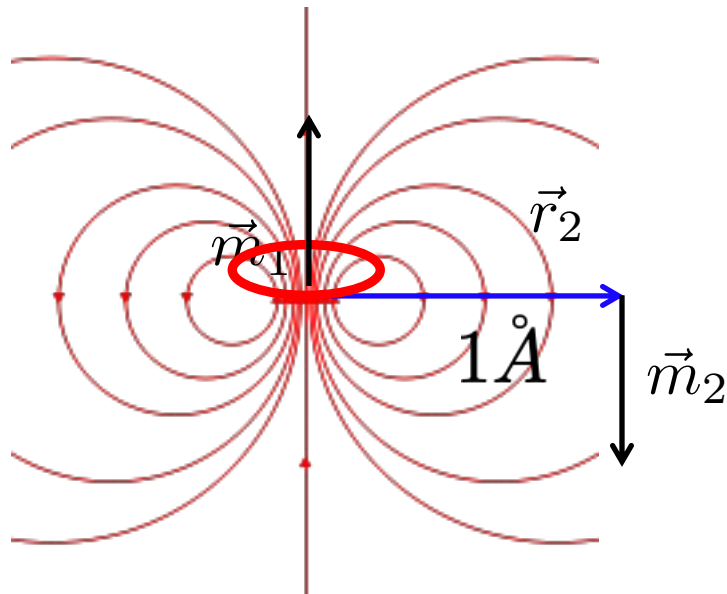
Facts that can not be explained:

- All “open shell “atoms have maximal spin
- Intra-atomic “spin energy” : $> 1\text{eV}$

$$\vec{m}_1 = \vec{m}_2 = \mu_B(0, 0, 1)$$

$$\Delta U \simeq 0.12 \text{ meV}$$

Other ‘magnetic glue’ must exist



$$\vec{m}_1 = \vec{m}_2 = \mu_B(0, 0, 1)$$

$$\Delta U \simeq 0.12 \text{ meV}$$

Magnetic Dipolar interaction is 10^5 times weaker than “intra-atomic” spin energy (0.06 meV vs 3.26 eV)

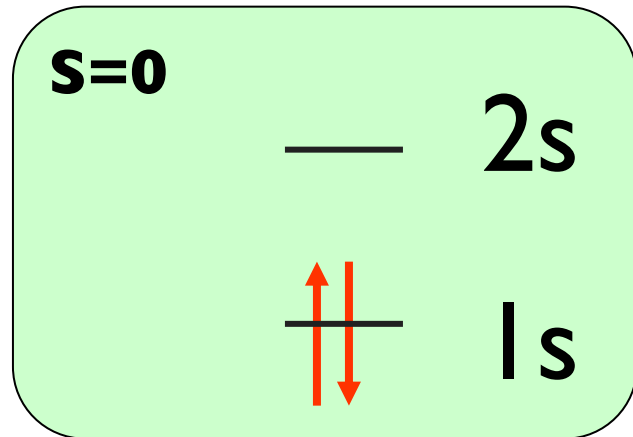
- Intra-atomic “spin energy” : $> 1 \text{ eV}$

Energy levels for Mn^{2+}

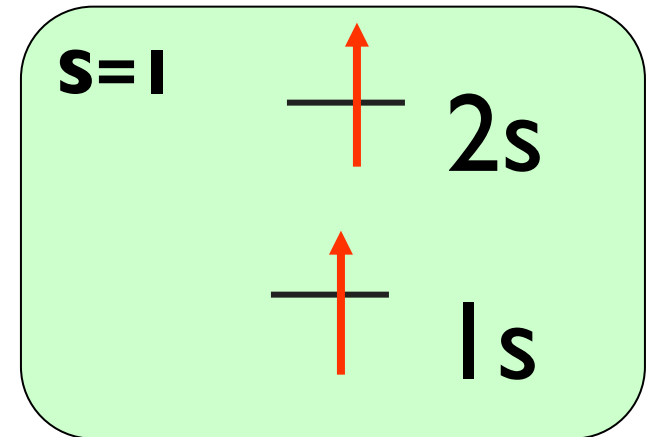
Configuration	Term	J	Level (eV)
$3d^5$	a^6S	$5/2$	0.000000
$3d^5$	a^4G	$11/2$	3.325802

https://physics.nist.gov/PhysRefData/ASD/levels_form.html

Closed shell vs Open shell

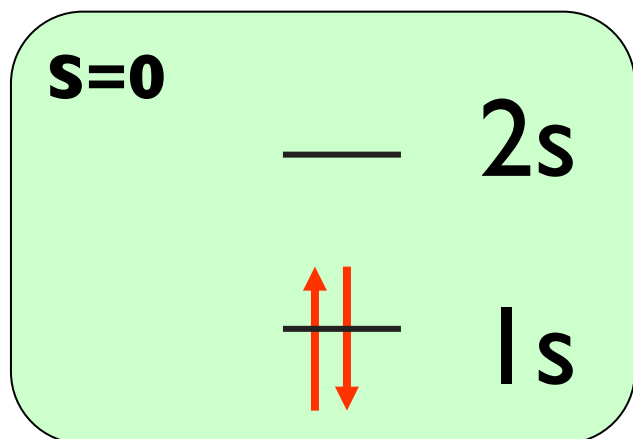


Ground state

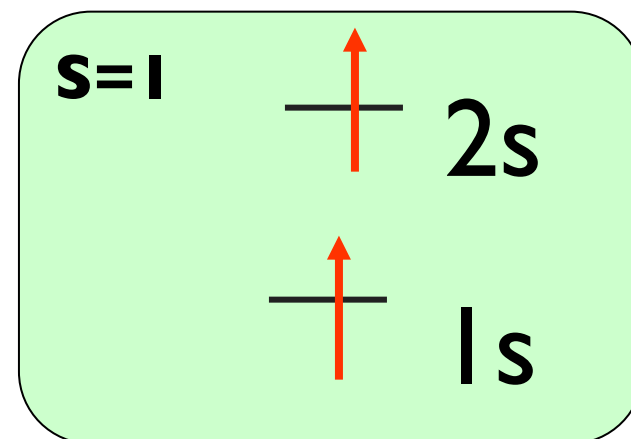
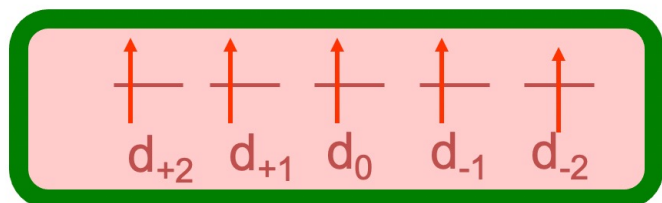


Excited state
(change of "single" particle energy)

Closed shell vs Open shell



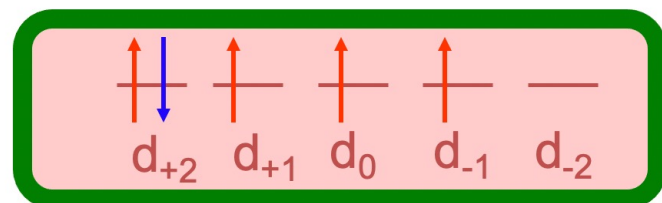
Ground state

Excited state
(change of "single" particle energy)

$$S=5/2$$

$$L=0$$

$$J=5/2$$



$$S=3/2$$

$$L=4$$

$$J=11/2$$

Configuration	Term	J	Level (eV)
$3d^5$	a^6S	$5/2$	0.000000
$3d^5$	a^4G	$11/2$	3.325802

https://physics.nist.gov/PhysRefData/ASD/levels_form.html

First Hund Rule

The term with maximum spin multiplicity (S) lies lowest in energy

Second Hund Rule (*)

For a given multiplicity (spin), the term with the largest value of L (angular momentum) has the lowest energy.

Third Hund Rule (spin orbit)

$$J = |L - S| \text{ if } n < 2l + 1$$

$$J = L + S \text{ if } n > 2l + 1$$



Friedrich Hund

Magnetic atoms

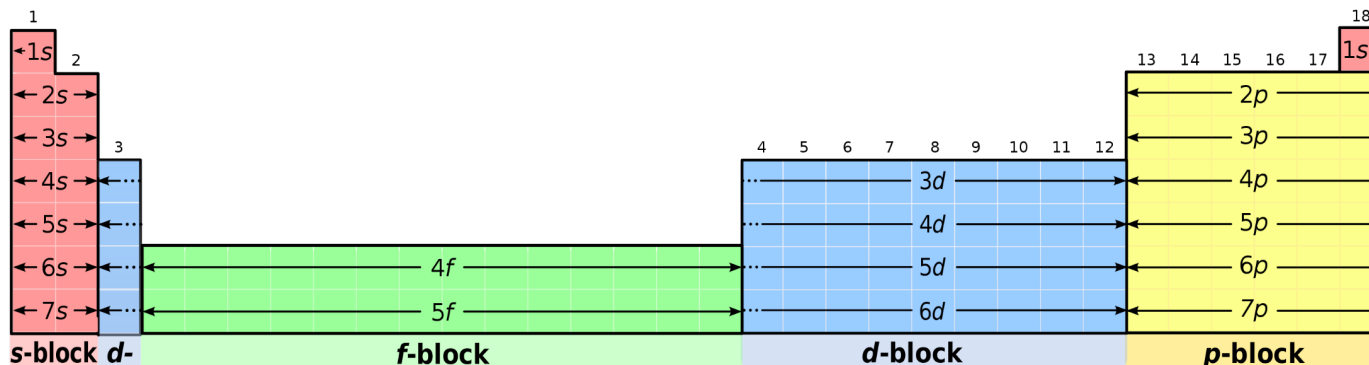
$$\vec{m} = \mu_B \left(g\vec{S} + \vec{L} \right)$$

- Finite spin requires OPEN SHELL configurations
- Intra-atomic Coulomb repulsion favours large S and large L
- Spin is robust, L is fragile
- (The dominant contribution to) Magnetism comes from spins

Intra-atomic exchange wrap up

Intra-atomic exchange mechanism

- 1) Open shell atoms with more than 1 electron in the shell



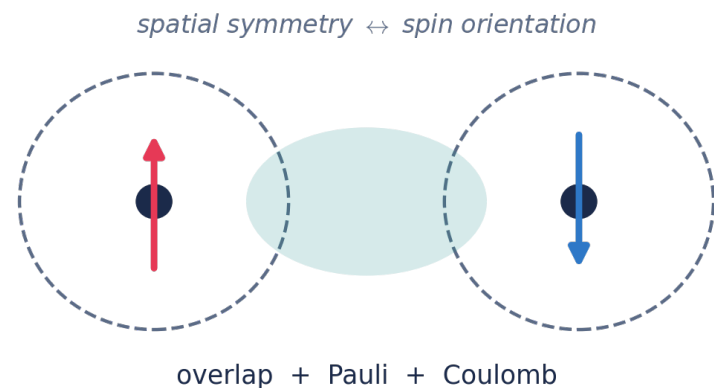
- 2) Cr, Fe, Co, Ni, Gd retain their magnetic moment in crystal phases.
- 3) 3d ions (Ti, V, Cr, Mn, Fe, Ni, Co, Cu), rare earth retain their magnetic moment in **many compounds** (oxides, halides, chalcogenides)

Exchange: Coulomb in disguise

Two electrons, the Heitler–London picture

The Pauli principle ties the spatial wavefunction's symmetry to the spin state. Symmetric vs antisymmetric spatial parts have different Coulomb energy — so the electrostatic energy ends up depending on relative spin orientation.

$$E_S - E_T = 2J, \quad J = \left\langle \phi_a \phi_b \left| \frac{e^2}{r_{12}} \right| \phi_b \phi_a \right\rangle$$



The magnetic interaction is electrostatic — enforced by the exclusion principle. No magnetic force required.

Traditionally three kinds of magnetic order

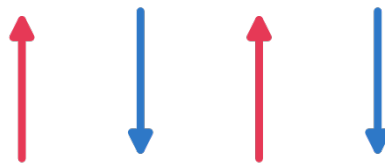
Ferromagnet



$$M \neq 0$$

Moments parallel on every site; large net M below T_C

Antiferromagnet



$$M = 0$$

Two sublattices cancel; zero net M but real order

Ferrimagnet

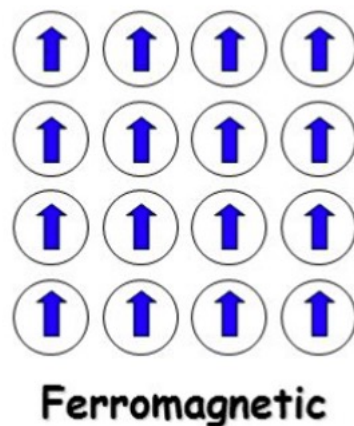
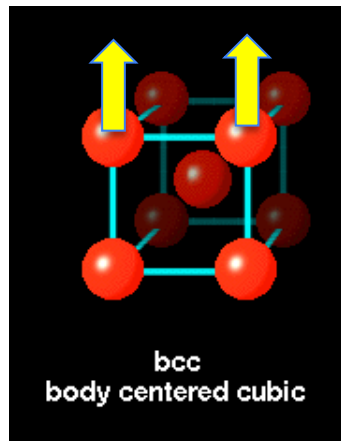


$$M \neq 0$$

Opposed but unequal sublattices; net M survives

Order parameter: the sublattice magnetization, nonzero only below the ordering temperature.

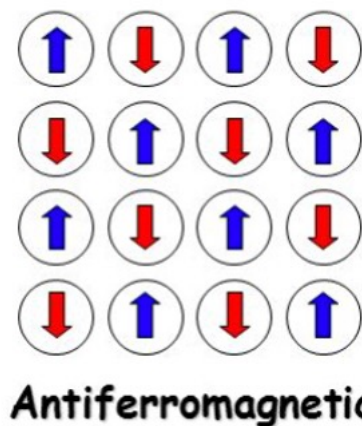
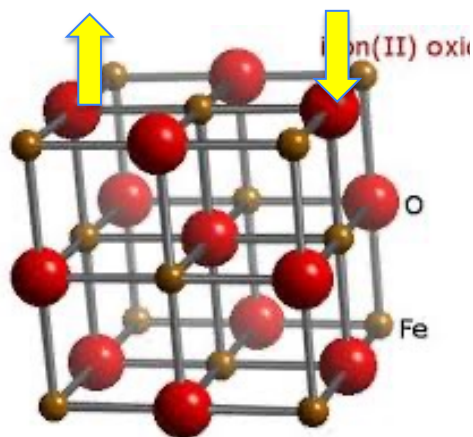
Magnetic order as evidence of interatomic exchange



Order parameter

$$\vec{M} = \sum_n \vec{m}(n)$$

Magnetization

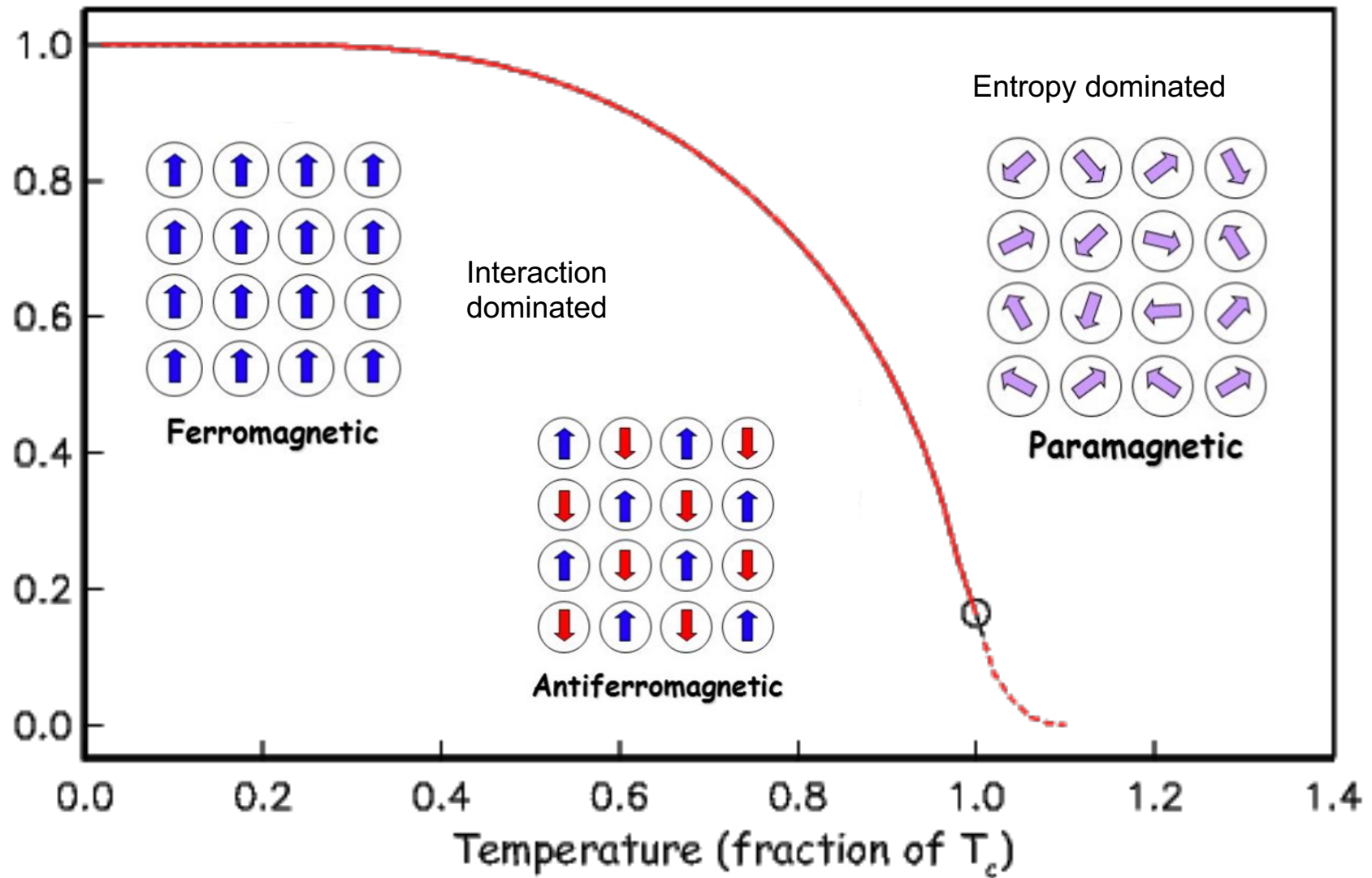


Order parameter

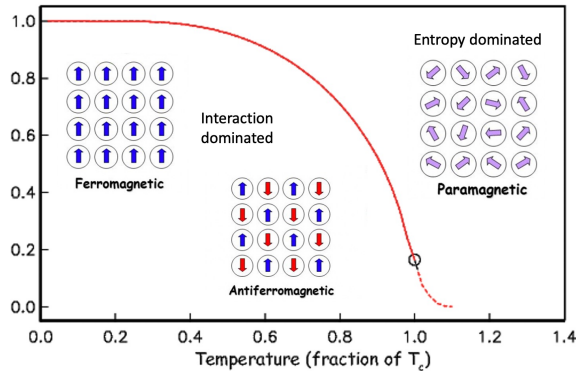
$$\vec{N} = \sum_{n \in A} \vec{m}(n) - \sum_{n' \in B} \vec{m}(n')$$

Staggered magnetization

Magnetic order as evidence of interatomic exchange



Magnetic order as evidence of interatomic exchange



$$k_b T_C = \frac{S(S+1)}{3} Jz$$

Mean field theory
Formula for Heisenberg
model

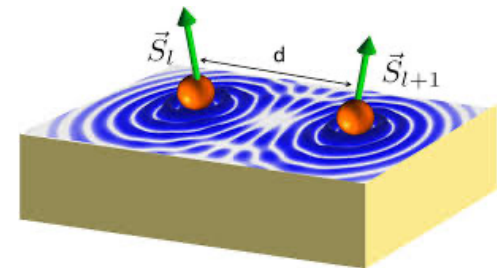
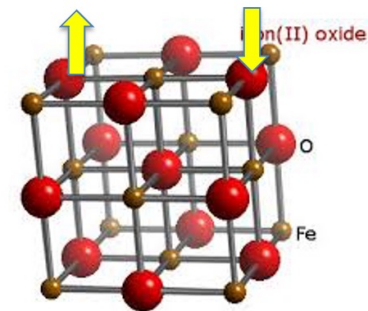
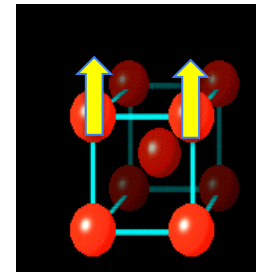
Material	Curie temperature (K)
Fe	1043
Co	1388
Ni	627
Gd	293
Dy	85
CrBr ₃	37
Au ₂ MnAl	200

Substance ♦	Néel temperature (K) ♦
MnO	116
MnS	160
MnTe	307
MnF ₂	67
FeF ₂	79
FeCl ₂	24
FeI ₂	9
FeO	198
FeOCl	80
NiO	525

Inter atomic exchange mechanisms

$$\mathcal{H} = J(\vec{r}_1 - \vec{r}_2) \vec{S}_1 \cdot \vec{S}_2$$

- Ferromagnetic **direct exchange**: (Fe-Fe)
(similar to intra-atomic exchange)
- **Superexchange**
 - Antiferromagnetic (Fe-O-Fe) linear bonds in FeO
 - Ferromagnetic Cr-I-Cr in CrI₃
- **Indirect exchange** (RKKY) in diluted magnetic alloys (oscillatory behaviour with distance)



The Heisenberg model & the sign of J

$$\hat{\mathcal{H}} = -2 \sum_{\langle i, j \rangle} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

J > 0 Ferromagnetic

Parallel spins favoured.
Direct exchange between
overlapping orbitals tends
this way.

J < 0 Antiferromagnetic

Antiparallel favoured. 180°
superexchange through a
bridging ligand is the classic
case.

 Conventions differ (2J, -J, +J). Pick one and stay consistent.

Key word in quantum science

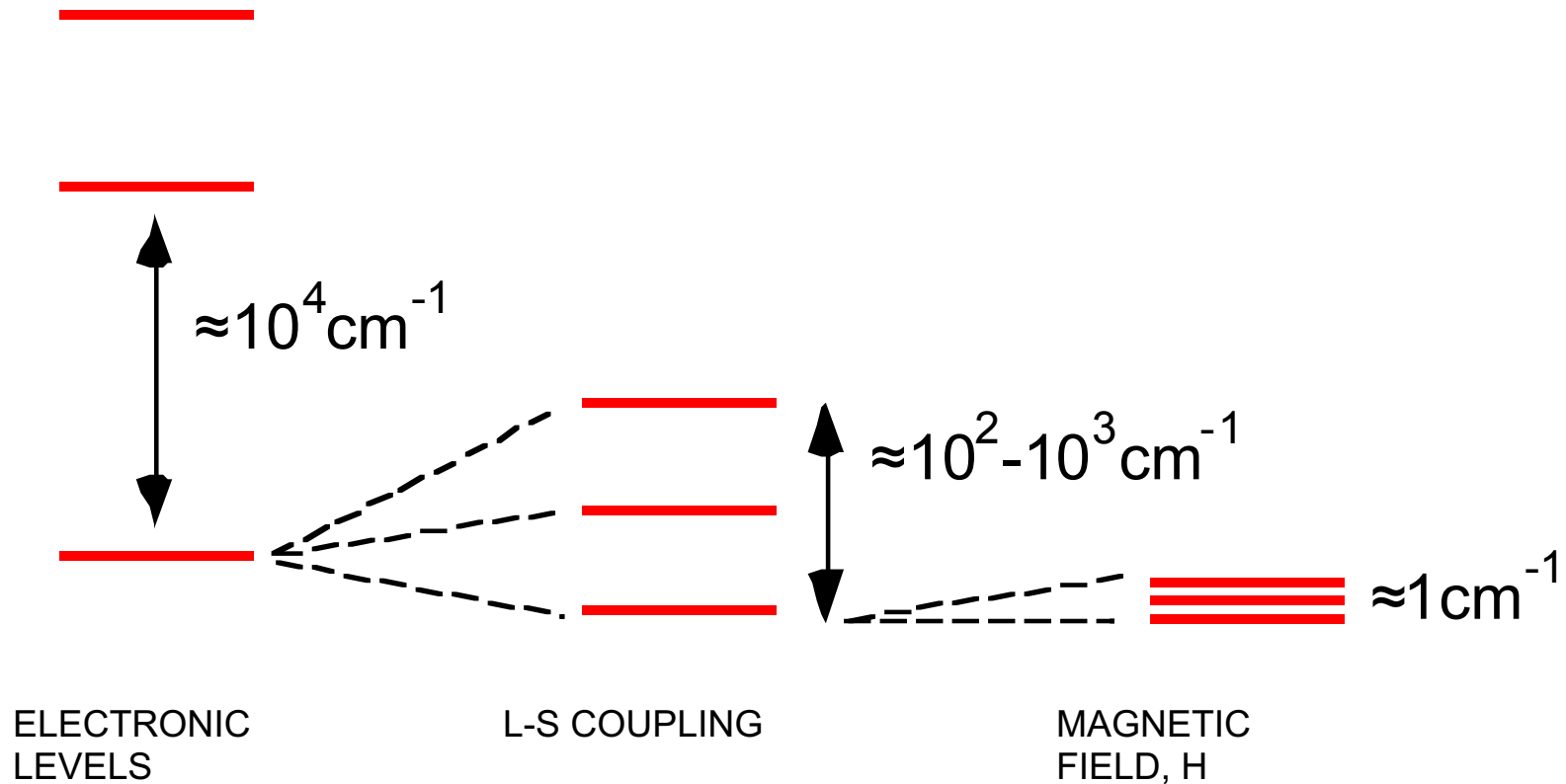
Interaction-Coupling

- Electron-nucleus
- Electron-electron
- Spin-Orbit (LS)
- Spin-Spin (S_1S_2) MAGNETIC INTERACTION
- Spin-Magnetic field (SH) ZEEMAN EFFECT

Consequence

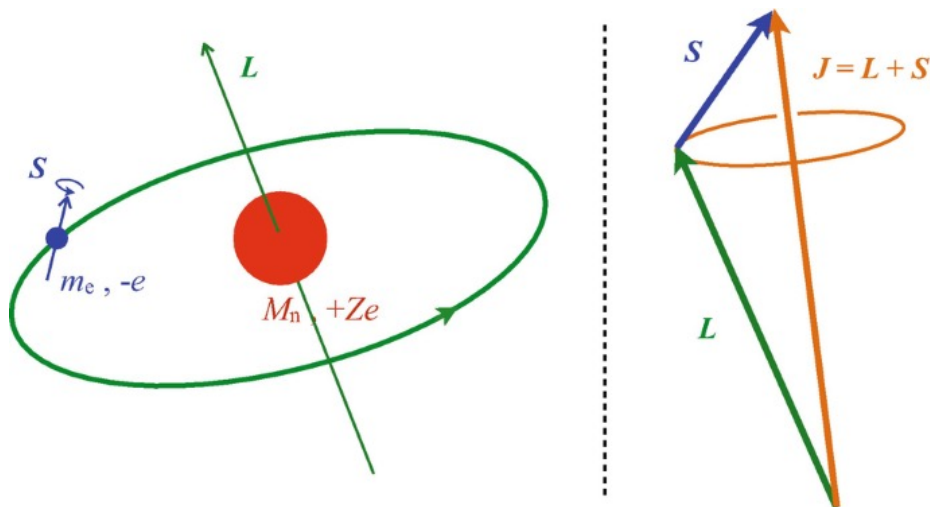
Energy splitting

Energy scales involved in magnetism



MAGNETISM: Study of the ground state

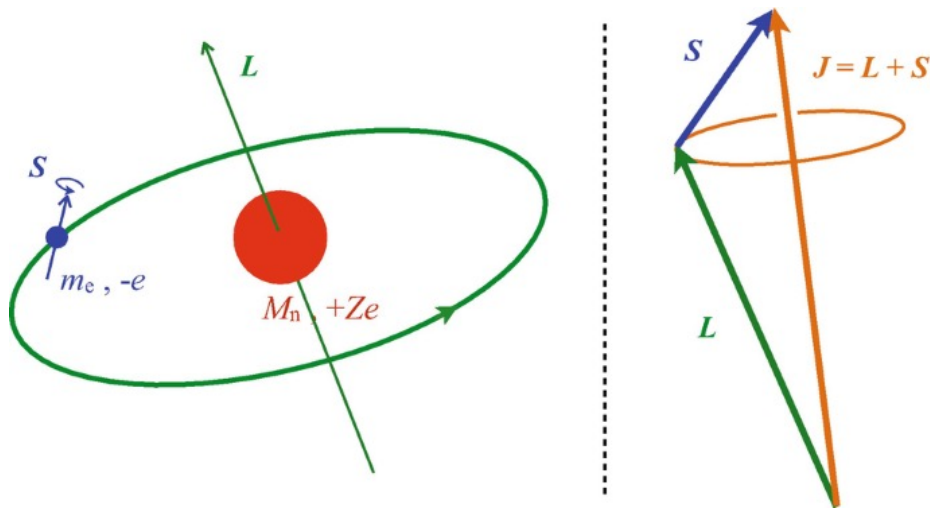
Spin orbit interaction



$$\mathcal{H}_{\text{SO}} = \lambda \vec{S} \cdot \vec{L}$$

- The strength of spin orbit coupling (SOC) depends on nuclear charge (Z^2)
- It can be as high as 1eV (for heavy atoms Pt, Bi)
- For Mn, Fe, Co, Ni is in the 10-20 meV range

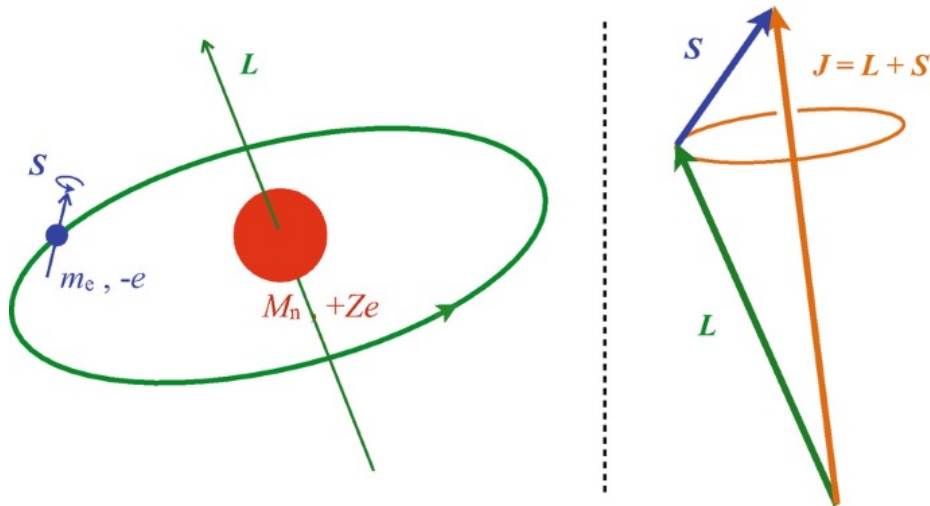
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- For atoms in solids, the effect is weaker but very important

Spin orbit interaction



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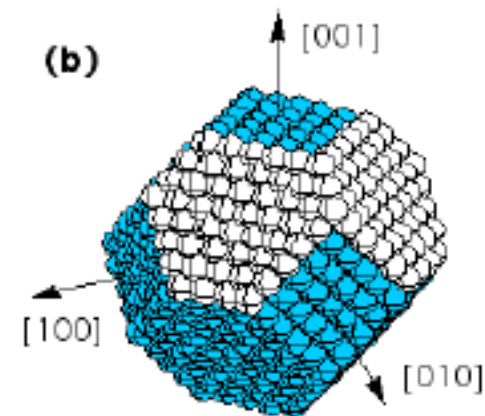
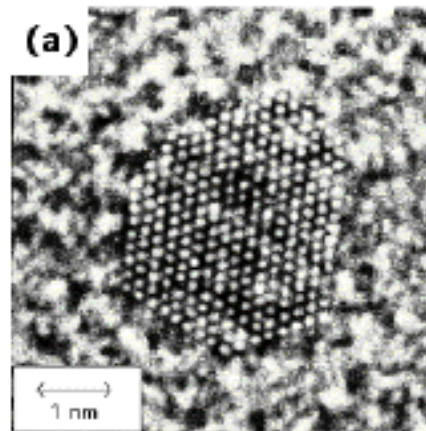
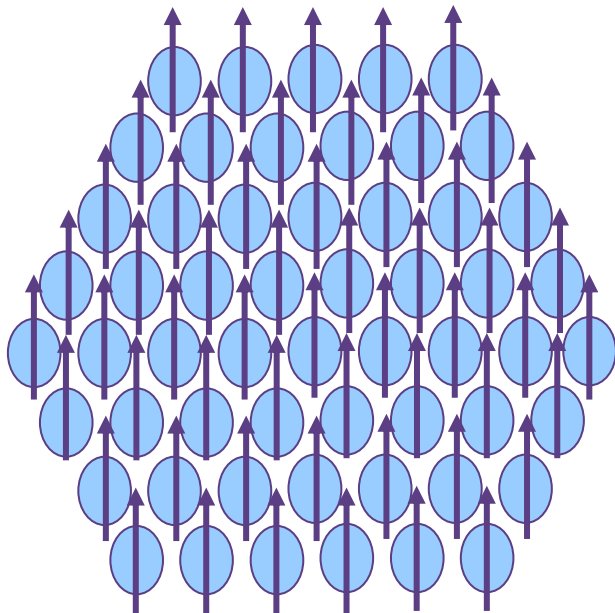
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- In free atoms, L is not quenched: very strong effect on energy levels
- For atoms in solids, the effect is weaker but very important
- SOC connects the abstract spin space to the orbital degrees of freedom
- Symmetry of crystal field (CF) determines the shape of the orbitals
- SOC + CF result in **magnetic anisotropy**

Magnetic anisotropy

The energy of a magnetic system depends on the orientation of the magnetic Moment, in the absence of an applied magnetic field.

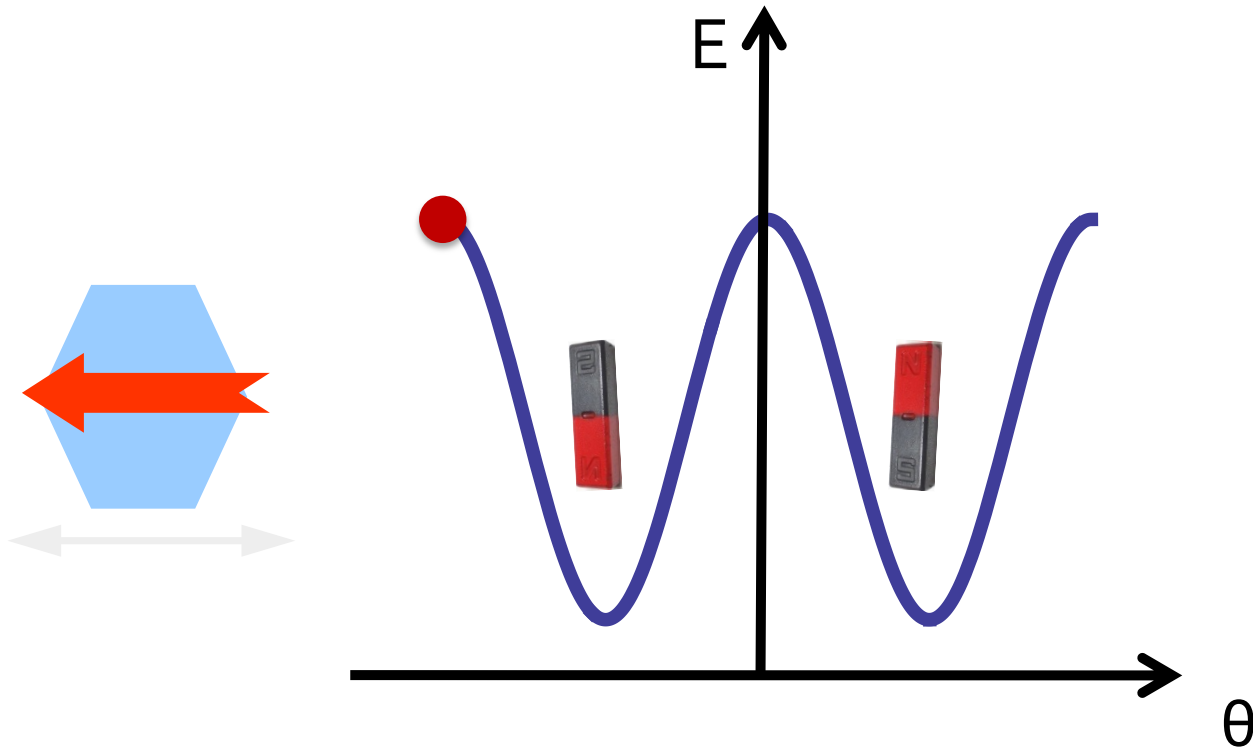
Relevant for:

- Bulk ferromagnets
- Nanomagnets
- Magnetic molecules
- Single atoms on surfaces



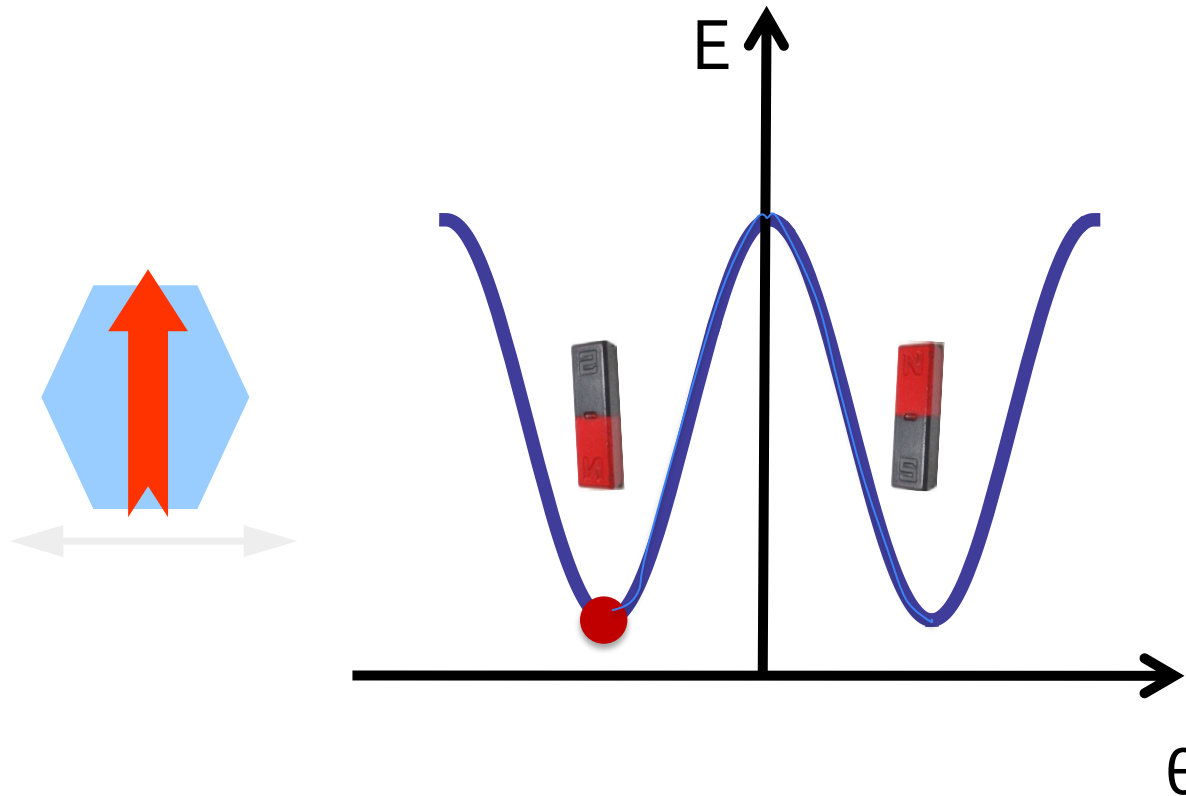
Uniaxial magnetic anisotropy: classical model

$$U = DM_z^2 = K \cos^2(\theta)$$



Uniaxial magnetic anisotropy: classical model

$$U = DM_z^2 = K \cos^2(\theta)$$



Anisotropy: why direction matters

Exchange sets whether moments align; anisotropy sets which way they point. Without it, there is no permanent magnet.

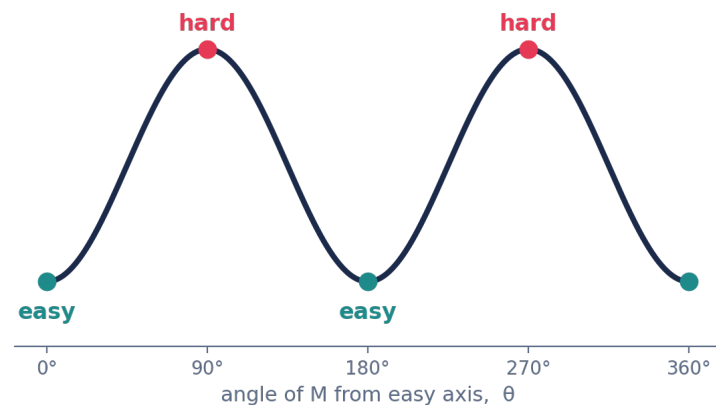
Magnetocrystalline anisotropy

Spin-orbit coupling ties the spin to the lattice, so some crystal directions cost less energy than others. To lowest order:

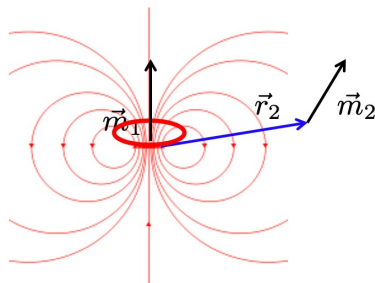
$$E_a = K_1 \sin^2 \theta$$

K_1 ranges from $\sim 10^2 \text{ J m}^{-3}$ (soft iron) to $\sim 10^7 \text{ J m}^{-3}$ ($\text{Nd}_2\text{Fe}_{14}\text{B}$). The spread between transformer cores and rare-earth magnets.

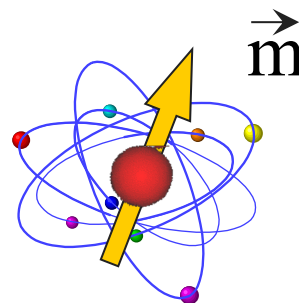
Energy landscape $E(\theta)$



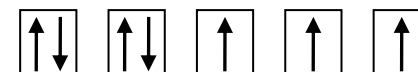
Easy axis ($\theta = 0$): energy minimum. Hard axis ($\theta = 90^\circ$): maximum.

Dipolar Interaction:

- Very Weak (μeV) and Long range
- Responsible of “shape anisotropy”

Intra-atomic exchange,

d or *f* electrons

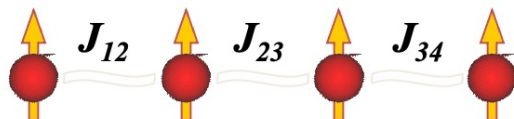


Hund's rules

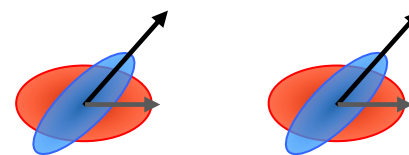
- Very Strong (3 eV) and short range
- Responsible of atomic moments

Inter-atomic exchange.

$$H_{exc} = -\sum_{i \neq j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$$



- Strong (up to 20-30 meV)
- Short range
- Responsible of magnetic order

Spin-Orbit Coupling:

$$H_{s.o.} = \lambda \mathbf{L} \cdot \mathbf{S}$$

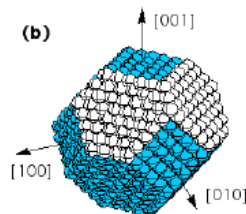
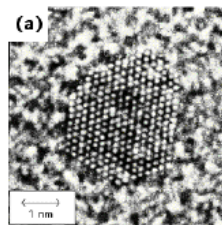
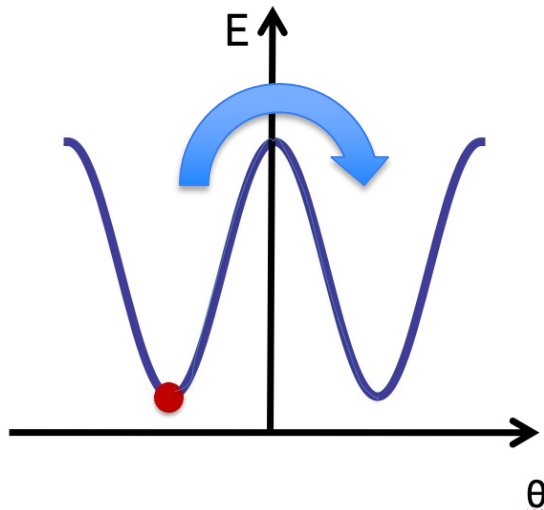
- Very Strong (1 eV) and short range
- Responsible of magnetic anisotropy

Superparamagnetism

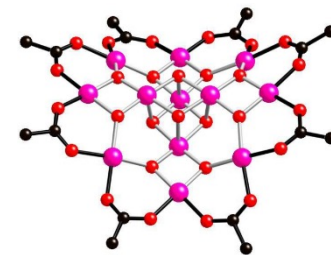
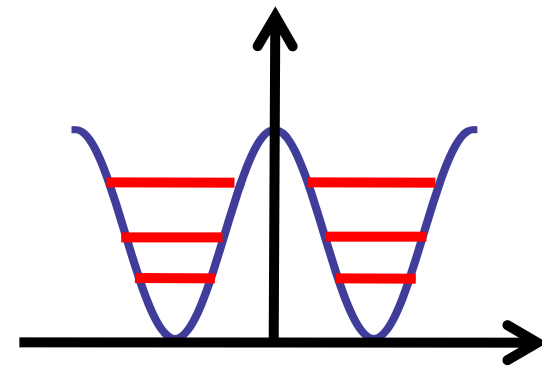
- Phenomenon that affects to monodomain magnets
- Monodomain -> nanomagnets
- Thermal flips of magnetization, destroying remanence
- Obstacle for miniaturization in magnetic recording

Anisotropy and energy landscape

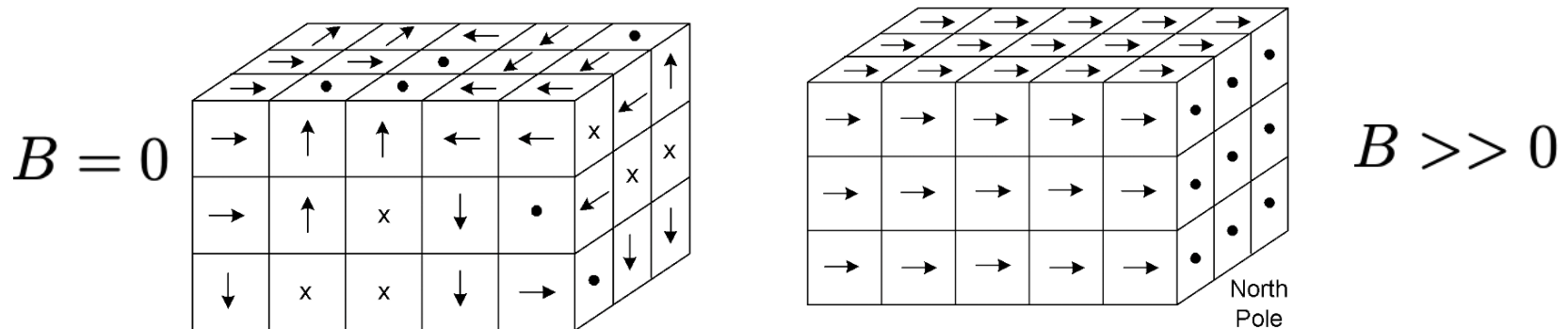
Thermal fluctuations:
Super-paramagnetism



Quantum spin tunneling

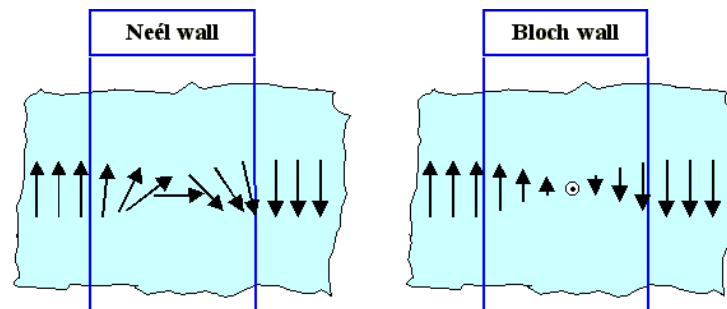


Magnetic domains

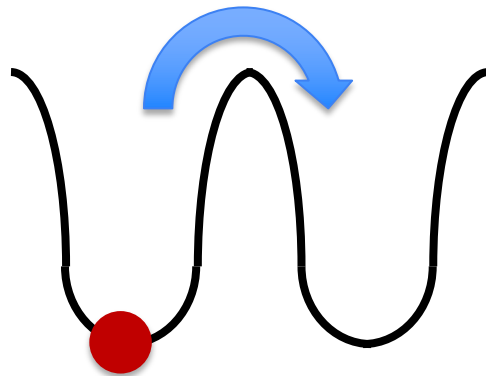


What is the size of the magnetic domains?

- Exchange and Zeeman favor monodomain state
- Dipolar interactions favors multi-domain state with perpendicular alignment
- Domain walls cost exchange and anisotropy energy



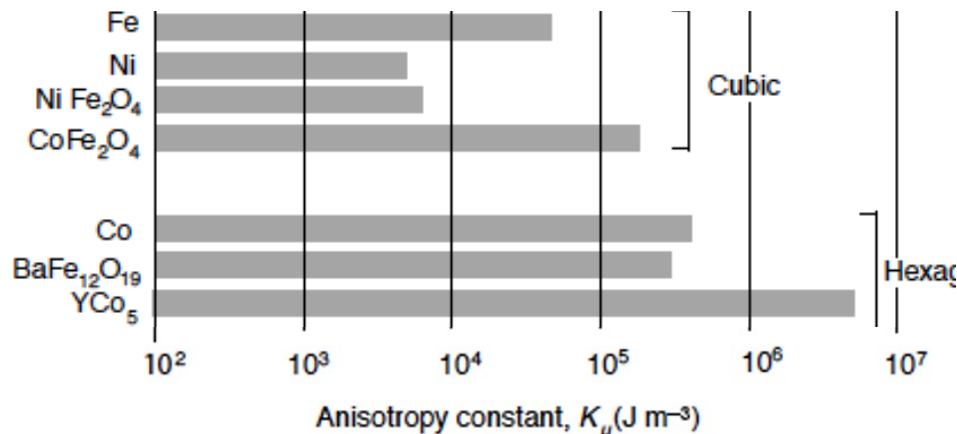
Magnetic domains



$$\tau_N = \tau_0 \exp\left(\frac{KV}{k_B T}\right)$$

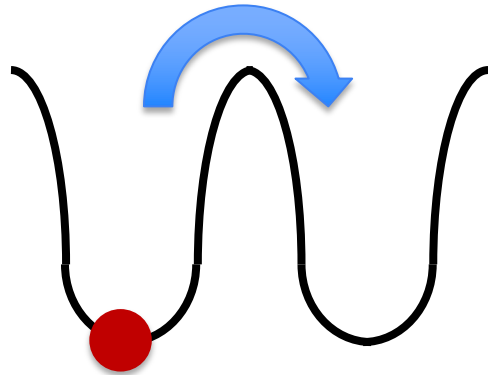
K MAE per unit volume

$$\tau_0 \simeq 10^{-9} \text{ s} \quad \text{Attempt time}$$



Cobalt MAE
1 micro eV per atom

Magnetic domains



$$\tau_N = \tau_0 \exp\left(\frac{KV}{k_B T}\right)$$

K MAE per unit volume

$$\tau_0 \simeq 10^{-9} \text{ s} \quad \text{Attempt time}$$

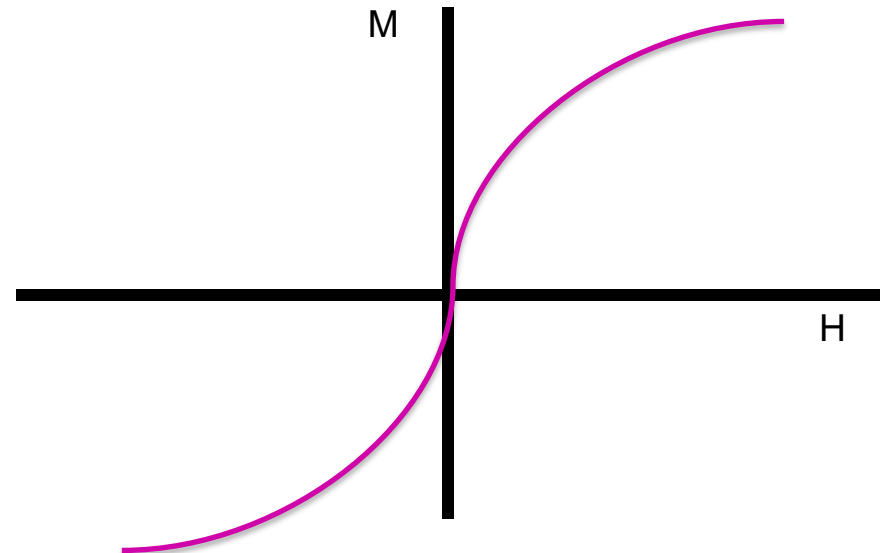
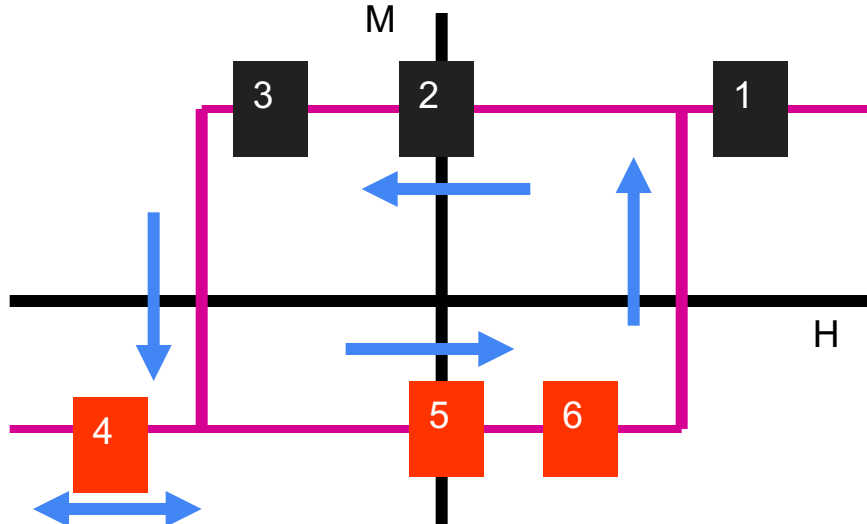
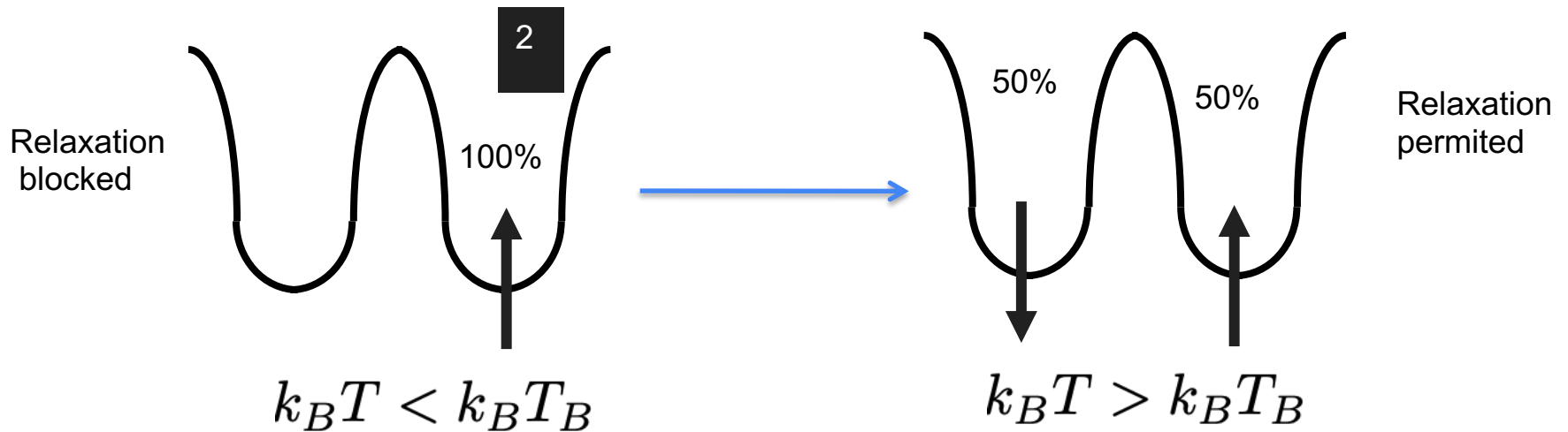
$N=10^6$ Cobalt atoms
 $KV= 10^6 \text{ atoms } 10^{-6} \text{ eV}=1 \text{ eV}$
 $k_B T= 26 \text{ meV}$
 Neel time=584 days

Exercise:

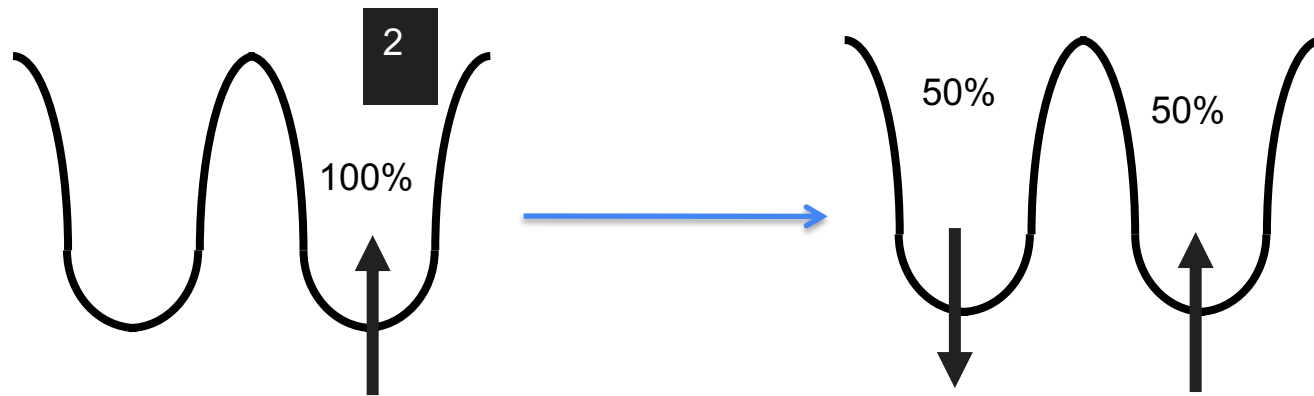
- compute Neel relaxation time for $N=5 \cdot 10^5$
- compute Neel relaxation time for $N=5 \cdot 10^6$

Cobalt MAE
 1 micro eV per atom

Magnetization curve: $T < T_B$ vs $T > T_B$

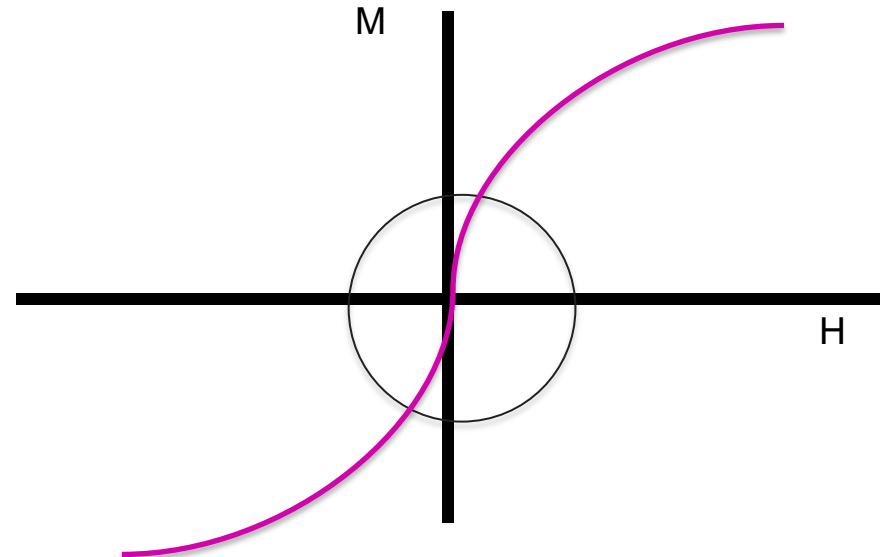
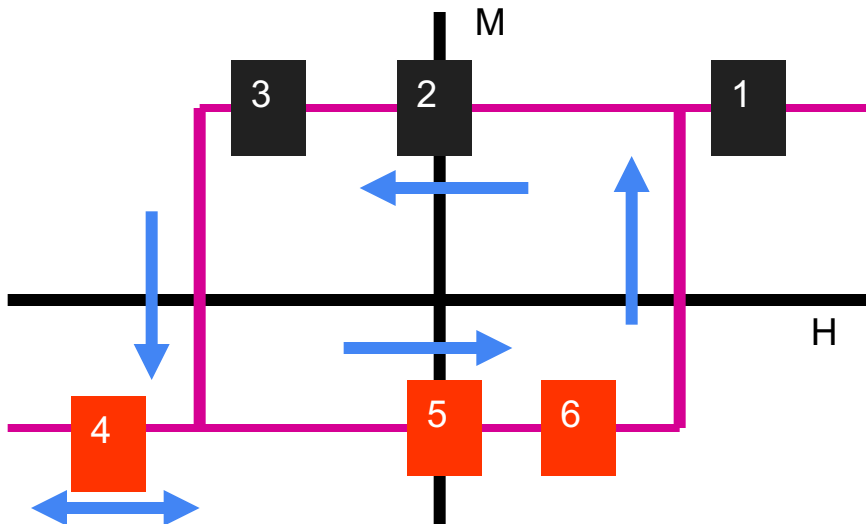


Magnetization curve: $T < T_B$ vs $T > T_B$

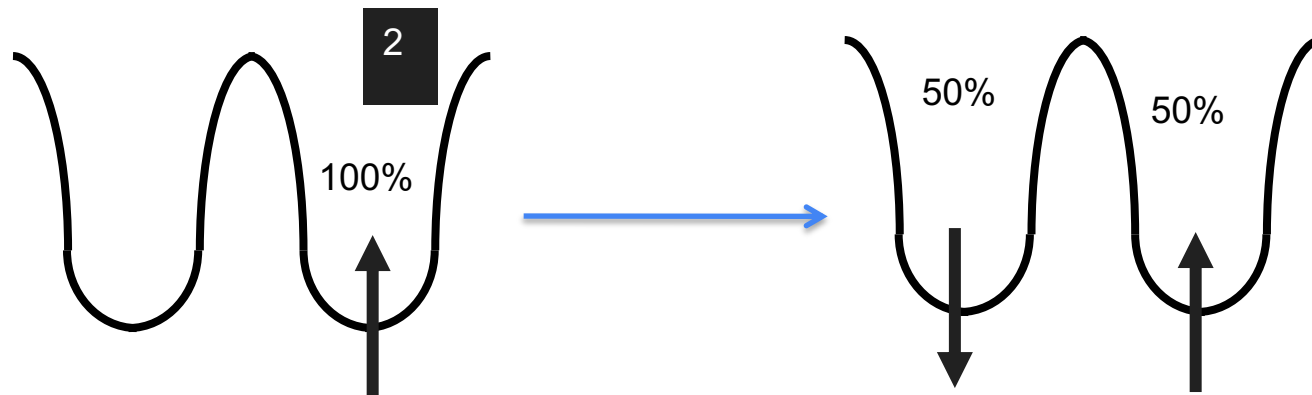


Relaxation blocked

Relaxation permitted

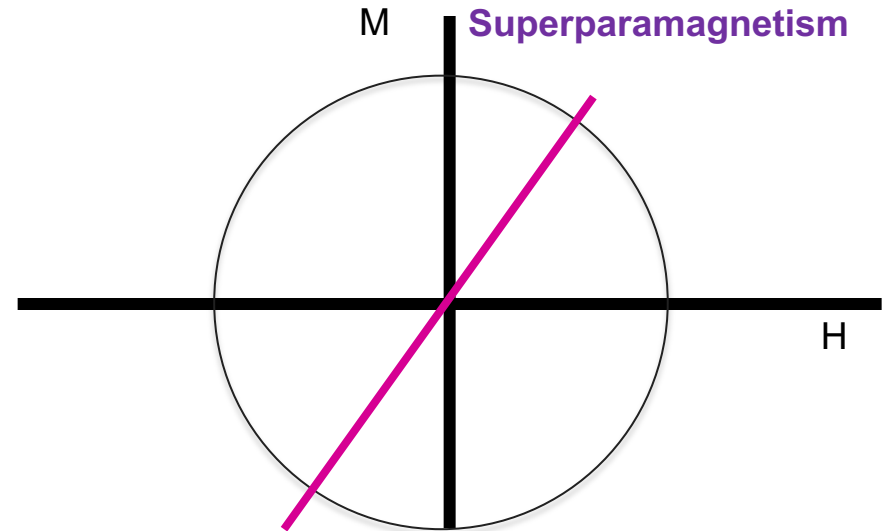
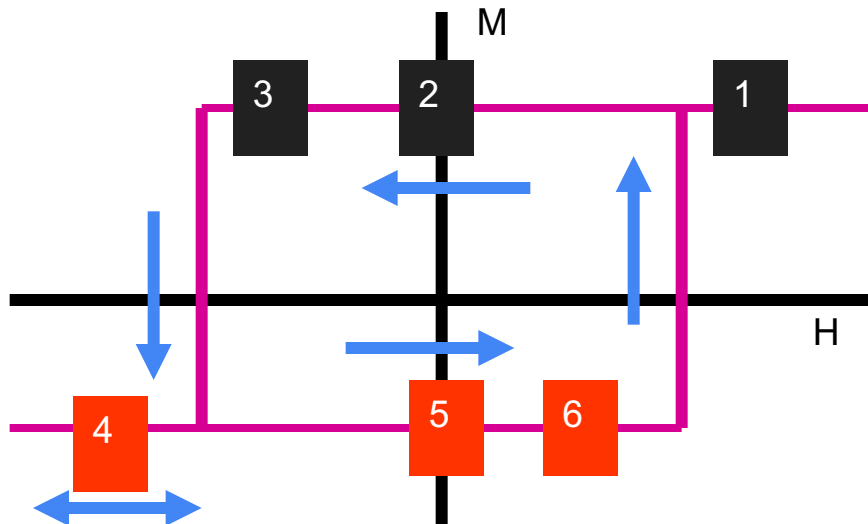


Magnetization curve: $T < T_B$ vs $T > T_B$



Relaxation blocked

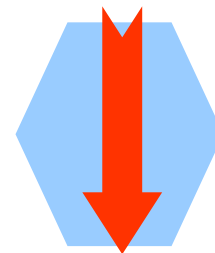
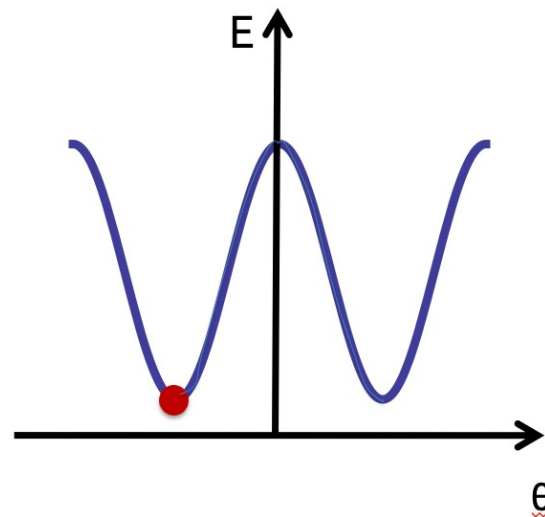
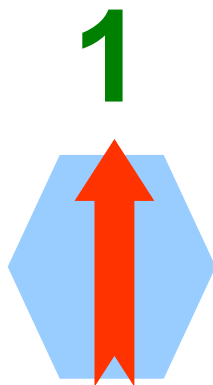
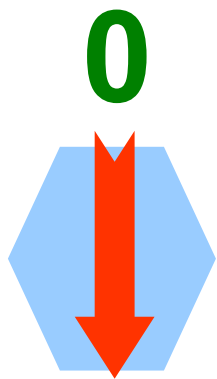
Relaxation permitted



Superparamagnetism: wrap up

- Affects nanomagnets
- Neel relaxation time < measurement time
- Random fluctuations of magnetization
- Linear $M(H)$ at small field: superparamagnetism
- Challenge for miniaturization in magnetic recording

$$\tau_N = \tau_0 \exp\left(\frac{KV}{k_B T}\right)$$



SYNTHESIS

What you can now reason about



With these four ideas you now have the tools to take on:

Domains & hysteresis

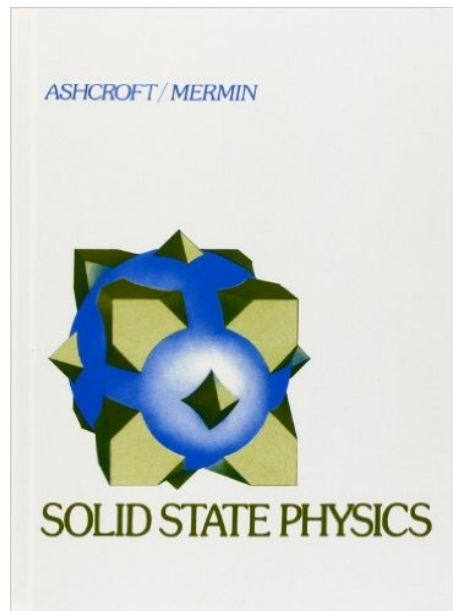
How real magnets minimise energy and remember their history.

Spin waves (magnons)

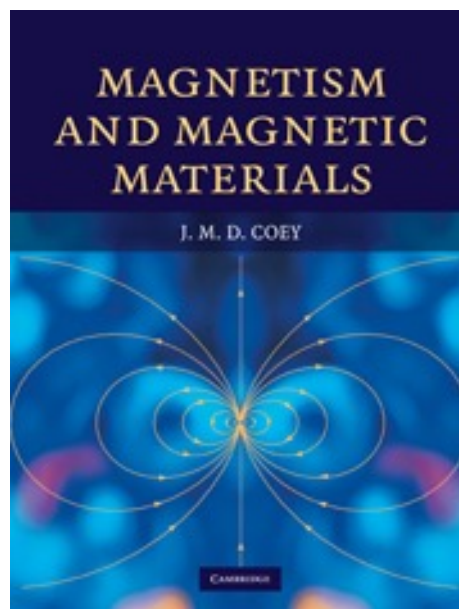
The low-energy excitations that mean field misses.

Frustration & disorder

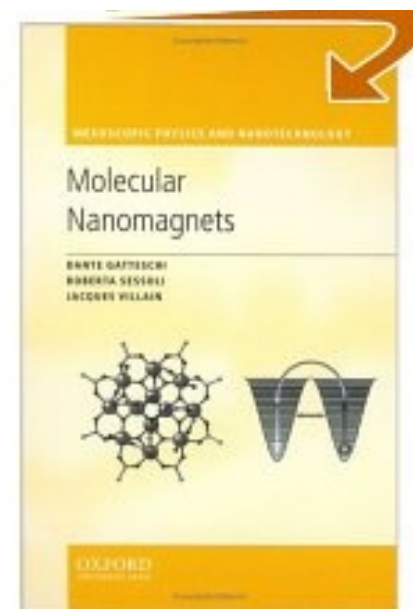
Spin glasses, frustration, and where order fails to form.



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and N. David
Mermin.
"Solid state
physics." (1976).



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MD. *Magnetism and
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Cambridge university
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